

§1 Conditional Optimization via Trigonometric Mappings

This section treats the verification of symmetric rational inequalities under cyclic boundary constraints by translating algebraic systems into convex trigonometric functions.

Problem Statement

Let a, b , and c be positive real numbers such that $a + b + c = abc$. Prove the following inequality:

$$\sqrt{1 + \frac{1}{a^2}} + \sqrt{1 + \frac{1}{b^2}} + \sqrt{1 + \frac{1}{c^2}} \geq 2\sqrt{3}$$

Source: International Mathematical Olympiad Variants

1.1 Rigorous Algebraic Proof

Proof. The cyclic system features a classic constraint equation: $a + b + c = abc$. This restriction matches a well-known trigonometric identity for the tangents of the internal angles of a triangle. Specifically, if A, B , and C denote the interior angles of an acute triangle $\triangle ABC$ such that $A + B + C = \pi$, then their tangents expand as follows:

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

Because a, b , and c are strictly positive real parameters, we can map them directly into this trigonometric domain without losing generality by setting:

$$a = \tan A, \quad b = \tan B, \quad c = \tan C$$

where each angle is located within the open first quadrant: $A, B, C \in (0, \frac{\pi}{2})$.

We can substitute this change of variables directly into the individual radical components of our target inequality. Simplifying the first term using fundamental trigonometric identities yields:

$$1 + \frac{1}{a^2} = 1 + \frac{1}{\tan^2 A} = 1 + \cot^2 A = \csc^2 A$$

Taking the positive square root for a first-quadrant angle shows that the radical expression simplifies directly down to a cosecant function:

$$\sqrt{1 + \frac{1}{a^2}} = \csc A = \frac{1}{\sin A}$$

Applying this transformation symmetrically across all three radical terms converts our original algebraic inequality into a clean, cyclic trigonometric expression:

$$\frac{1}{\sin A} + \frac{1}{\sin B} + \frac{1}{\sin C} \geq 2\sqrt{3}$$

To prove that this relationship always holds true, we can check the convexity of the underlying function $f(x) = \frac{1}{\sin x}$ over the domain interval $(0, \frac{\pi}{2})$. Calculating its derivative properties gives:

$$\begin{aligned} f'(x) &= -\csc x \cot x \\ f''(x) &= \csc x \cot^2 x + \csc^3 x \end{aligned}$$

Since both $\csc x$ and $\cot^2 x$ are strictly positive values for all angles between 0 and $\frac{\pi}{2}$, the second derivative $f''(x)$ remains strictly positive across our entire domain. This establishes that $f(x) = \frac{1}{\sin x}$ is a strictly convex function.

By invoking Jensen's Inequality for convex functions, we can assert that the average of the function values at these three points must be greater than or equal to the function evaluated at their arithmetic mean:

$$\frac{f(A) + f(B) + f(C)}{3} \geq f\left(\frac{A + B + C}{3}\right)$$

Substituting our specific cosecant relationships and using the structural fact that $A + B + C = \pi$ gives:

$$\frac{\frac{1}{\sin A} + \frac{1}{\sin B} + \frac{1}{\sin C}}{3} \geq \frac{1}{\sin\left(\frac{\pi}{3}\right)}$$

Evaluating the constant on the right side using the exact value $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$ allows us to calculate its reciprocal:

$$\frac{1}{\sin\left(\frac{\pi}{3}\right)} = \frac{2}{\sqrt{3}}$$

Multiplying both sides of our inequality by 3 completes the proof of our intermediate step:

$$\frac{1}{\sin A} + \frac{1}{\sin B} + \frac{1}{\sin C} \geq 3 \cdot \frac{2}{\sqrt{3}} = 2\sqrt{3}$$

Reverting back to our original algebraic coordinates a, b , and c , we arrive at the final intended inequality statement:

$$\sqrt{1 + \frac{1}{a^2}} + \sqrt{1 + \frac{1}{b^2}} + \sqrt{1 + \frac{1}{c^2}} \geq 2\sqrt{3}$$

The absolute minimum is achieved under complete symmetry when $A = B = C = \frac{\pi}{3}$, which corresponds to the point where $a = b = c = \tan\left(\frac{\pi}{3}\right) = \sqrt{3}$. $\square$